

Adroit Technologies

Protocol Driver

Name	Mitsubishi FX2 and FX3 Extended FX Protocol
DLL	MITFXS



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1. Overview

Adroit supports communication with the Mitsubishi FX PLC range, using the red Beijer RS-232/422 converter cable and other programming cables. MitFXS is designed to replace the existing MitsuFX and MitsuFXS drivers which are discontinued (see [Appendix A](#) for compatibility with deprecated drivers).

This driver also supports communication using Dial-Up Modem.

This driver also supports communication using "Serial over Ethernet" connection that wraps the serial FX extended protocol into the data portion so as to be usable by Ethernet to serial gateways, the protocol is not modified.

2. Wiring

2.1. Direct

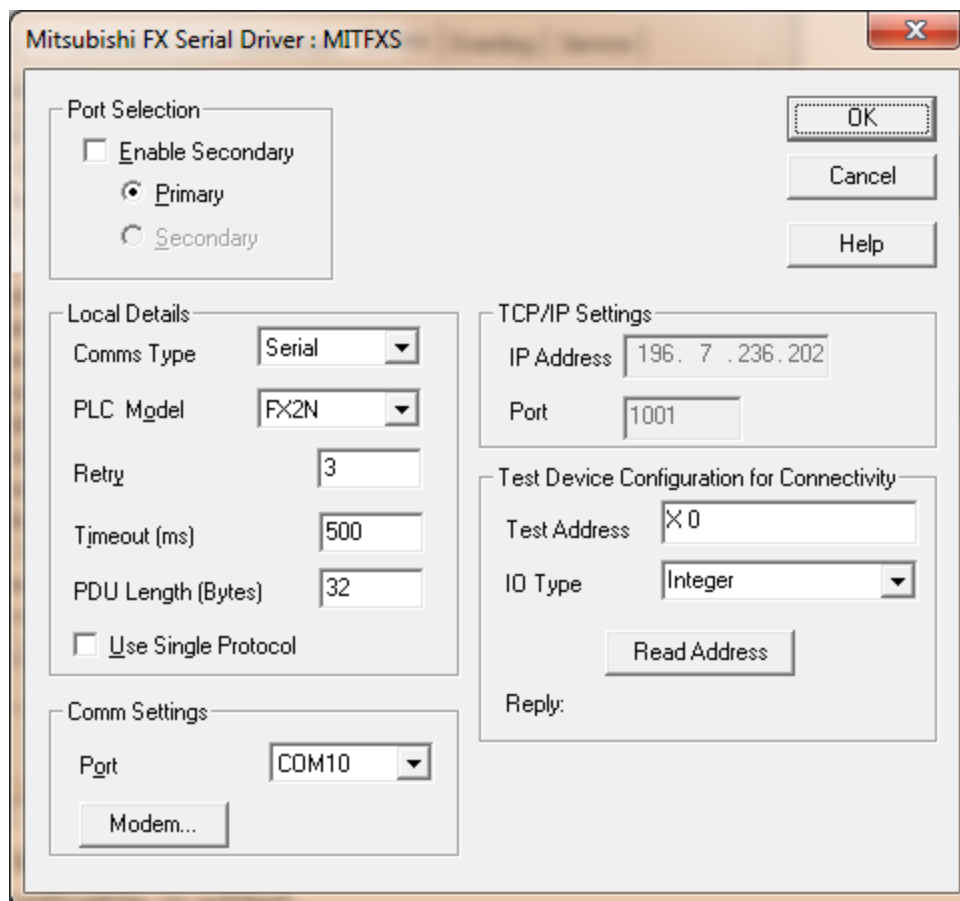
Connect using RS-232 programming cable from Mitsubishi. Red SC09 converter cable "RS232/RS422".

2.2. Modem

On the remote PLC side to connect the modem serial output to the SC09 cable, ensure that a crossover serial cable is used, because we want to double crossover i.e. MASTER TX to exit out of the MODEM TX line to enter the TX input of the SC09.

3. Device Configuration

3.1. Device Configuration



3.1.1. Port Selection

Enable Secondary – Enables secondary configuration for dual redundant PLCs. This setting is disabled for this driver.

Primary / Secondary – Select the primary or secondary button depending on which connection you wish to configure.

3.1.2. Local Details

Comms Type – Allows Selection of communication using Serial or Ethernet.

PLC Model - The various FX models. Valid I/O scan ranges are provided in the table below:

Model	Input **	Output **	Timers	Counters	Data	Auxiliary (M)	State	Special Auxiliary (MS)	Special Data (DS)
FX0	17	15	55	254	31	511	63	8255	8255
FX0s	17	15	55	254	31	511	63	8255	8255
FX0n	177	177	63	255	255	511	127	8255	8255
FX1s	17	15	63	255	255	511	127	8255	8255
FX1n	177	177	255	255	8255	1535	999	8255	8255
FX2n	267	267	255	255	8255	1535*	999	8255	8255

FX2nc	267	267	255	255	8255	1535*	999	8255	8255
FX3U	377	377	255	255	8255	3071	999	8255	8255

**Octal representation 177

Retry – The number of attempts that Adroit will try to communicate with a PLC before failing a transaction.

Timeout – The time that Adroit will wait for a valid response from a PLC before failing a transaction (Read or Write).

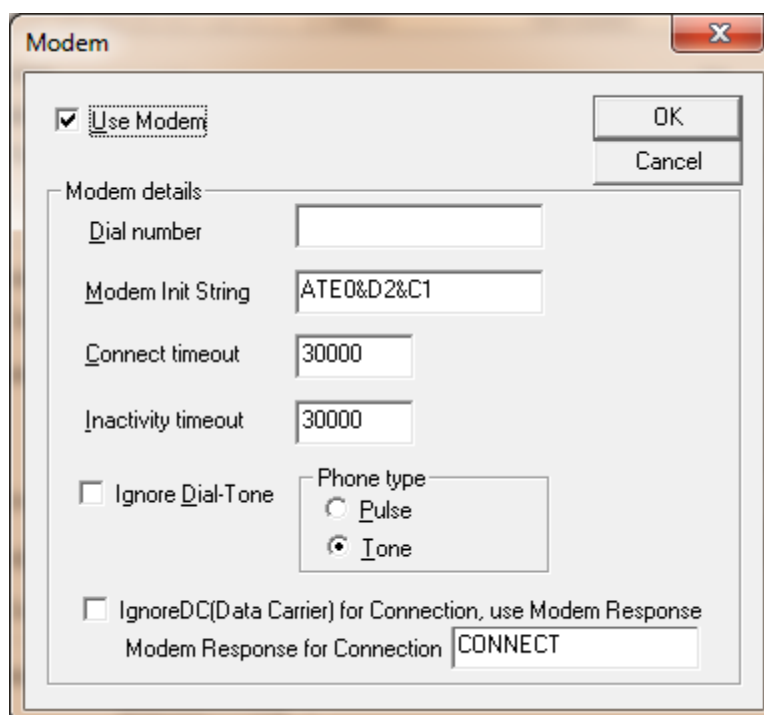
PDU Length – Indicates the maximum number of bytes that will be requested in a single Transaction from the PLC.

Use Single Protocol – This checkbox allows some of the protocol transactions to be combined and sent through as a single transaction, this is sometimes necessary when using modems. This checkbox is off by default and this option should only be used when necessary as the ACK/NAK/READY commands will be combined with other messages and sent as in a single transaction. This has the effect of not exceeding tight timing considerations for protocol such as the READY and ACK transactions, which would otherwise be delayed due to delays produced by the modems.

3.1.3. Comm Settings

Port – The available serial ports on your machine.

Modem – Allows Dial-Up Connection using Modems. Pressing this button will show the modem configuration window as shown below to setup modem to modem communications.



Use Modem – Enable/Disable Modem for communication.

Dial Number – Remote Modem number to dial.

Modem Init String – AT Command initialisation string that will be sent to the modem before dialling.

Connect Timeout – Maximum time to wait in milliseconds for a connection response from the remote modem.

Inactivity Timeout – Is the time to wait in milliseconds, before dropping the line due to no activity.

Ignore Dial Tone – Some telephone systems including GSM do not have dial tone.

Phone Type – Select Tone or Pulse Dialling (for very old switch exchanges).

IgnoreDC (Data Carrier) for Connection, use Modem Response – If no signalling is wired on the serial line between the master and its modem.

Modem Response for Connection – When ignoring data carrier, the response string that the driver can expect from the remote modem is defined here. Some modems reply with "CONNECT" others with "OK" etc.

3.1.4. TCP/IP Settings

IP Address – Destination TCP/IP address.

Port – Destination Port.

3.1.5. Test Device Configuration for Connectivity

This section is available to help users determine whether the settings they have entered are valid and Adroit can communicate with the device. The Agent Server is not required to be running.

Test Address – An address in your device which will be polled when the "Read Address" button is clicked. The address entered must be valid and will match the IEC or Standard Modbus addressing scheme selected in the advanced dialogue.

IO Type – Defines the base address type to be read, i.e. Boolean, Integer, Real or String80.

Read Address – Click this button to poll the PLC with the current settings in the dialog.

Reply – The successful or failed result will be updated to this static text control.

4. Supported Addresses

4.1. FX Data Types:

Device	Min Range	Max Range	PLC Data Type	Boolean	Integer	Real
Inputs X (octal)	0	*	Digital	X 0		
	0	*	Signed Integer		X 0 I	X 0 I
	0	*	Unsigned Integer		X 0 W	X 0 W
Outputs Y (octal)	0	*	Digital	Y 0		
	0	*	Signed Integer		Y 0 I	Y 0 I
	0	*	Unsigned Integer		Y 0 W	Y 0 W
Timers T	0	*	Signed Integer		T 0 I	T 0 I
	0	*	Unsigned Integer		T 0 W	T 0 W
Counters C (16 bit)	0	*	Signed Integer		C 0 I	C 0 I
		*	Unsigned Integer		C 0 W	C 0 W
Counters C (32 bit)	200	*	Signed Integer		C 0 D	C 0 D
Data D	0.0	*.15	Digital	D 0.0		
	0	*	Signed Integer		D 0 I	D 0 I
	0	*	Unsigned Integer		D 0 W	D 0 W
	0	*	Double word		D 0 D	D 0 D
	0	*	Float			D 0 F
Auxiliary M	0	*	Digital	M 0		
	0	*	Signed Integer		M 0 I	M 0 I
	0	*	Unsigned Integer		M 0 W	M 0 W
	0	*	Double word		M 0 D	M 0 D
Special S	0	*	Signed Integer		S 0 I	S 0 I
	0	*	Unsigned Integer		S 0 W	S 0 W
Special Auxiliary MS (read only)	8000	*	Digital	MS 8000		
	8000	*	Signed Integer		MS 8000 I	MS 8000 I
	8000	*	Unsigned Integer		MS 8000 W	MS 8000 W
	8000	*	Double word		MS 8000 D	MS 8000 D
Special Data Registers DS (read only)	8000	*	Digital	MS 8000		
	8000	*	Signed Integer		MS 8000 I	MS 8000 I
	8000	*	Unsigned Integer		MS 8000 W	MS 8000 W
	8000	*	Double word		MS 8000 D	MS 8000 D

- * Maximum range is determined by the PLC Model table defined in [Section 4.1.2. Local Details](#).
- I/O X and Y words are numbered by byte. I.e. Y0 (integer) includes all digitals from Y 0 up to Y17. (Count 16 bits). Y20 thus includes bits Y20 to Y37.
- The constant K setting of a timer is inaccessible via protocol, please use a D register for this purpose.

5. Protocol Definition and Considerations

Communications are fixed at 9600 baud, 7 data, 1 start, 1 stop and even parity. The protocol is essentially ASCII, with control codes delimiting portions of the messages. The LRC value is the 8-bit SUM of all bytes in a message from after the STX up until and including the ETX.

All PLC Types before the FX2N model uses the FX Protocol, the FX2N and FX3U use the Extended FX Protocol.

5.1. Using the FX Protocol

5.1.1. Read Register

Enquiry/ Ready?

ENQ
05
♣

Acknowledgement:

ACK
06
♠

Read request (Y0):

STX	CMD	Start Addr Hi	Start Addr Lo	Number bytes	ETX	LRC	LRC
02	30	30 30	30 3A	30 32	03	36	36
☺	0	0 1	0 0	0 2	♥	5	4

Read response:

STX	Data	ETX	LRC	LRC
02	xx xx xx xx	03	xx	xx
☺	X X X X	♥	X	X

5.1.2. Write Register

Enquiry/ Ready?

ENQ
05
♣

Acknowledgement:

ACK
06
♠

Write request (Y0):

STX	CMD	Start Addr Hi	Start Addr Lo	Number bytes	Data	ETX	LRC	LRC
02	31	30 30	30 3A	30 32	Xx xx xx xx	03	36	36

Write response:

ACK
06

5.1.3. Write Bit / Device Forcing

Enquiry/ Ready?

ENQ
05
♣

Acknowledgement:

ACK
06
♠

Write request (Y23): *force on*

STX	CMD	Start Addr Hi	Start Addr Lo	Number bytes	ETX	LRC	LRC
02	37	31 33	30 35	00 02	03	xx	xx

OR

Write request (Y23): *force off*

STX	CMD	Start Addr Hi	Start Addr Lo	Number bytes	ETX	LRC	LRC
02	38	31 32	30 35	00 02	03	xx	Xx

Write response:

ACK
♠
06

5.2. Using the Extended FX Protocol

5.2.1. Read Register

Enquiry/ Ready?

ENQ
05
♣

Acknowledgement:

ACK
06
♠

Read request (Y0):

STX	Extended	CMD	Start Addr Hi	Start Addr Lo	Number bytes	ETX	LRC	LRC
02	45	30	30 30 30	30 31	30 32	03	36	36
☺	E	0	0 0 0	0 1	0 2	♥	5	4

Read response:

STX	Data	ETX	LRC	LRC
02	xx xx xx xx	03	xx	xx
☺	X X X X	♥	X	X

5.2.2. Write Register

Enquiry/ Ready?

ENQ
05
♣

Acknowledgement:

ACK
06
♠

Write request (Y0):

STX	Extended	CMD	Start Addr Hi	Start Addr Lo	Number bytes	Data	ETX	LRC	LRC
02	45	31	30 30 30	30 31	30 32	Xx xx xx xx	03	36	36

Write response:

ACK
06

5.2.3. Write Bit / Device Forcing

Enquiry/ Ready?

ENQ
05
♣

Acknowledgement:

ACK
06
♠

Write request (Y23): **force on**

STX	Extended	CMD	Start Addr Hi	Start Addr Lo	Number bytes	ETX	LRC	LRC
02	45	37	30 31 33	30 35	00 02	03	xx	xx

OR

Write request (Y23): **force off**

STX	Extended	CMD	Start Addr Hi	Start Addr Lo	Number bytes	ETX	LRC	LRC
02	45	38	31 32	30 35	00 02	03	xx	Xx

Write response:

ACK
♠
06

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. Troubleshooting

It is possible that by modifying the default settings, that the **watchdog** sent by the driver may not arrive in the PLC, thus causing the PLC to “intermittently” reset our connection to the PLC. Ensure that the timer in the PLC (which runs for 120 seconds by default) is at least double the setting used by the driver. If the driver channel is backlogged (i.e. the driver is too busy with polling tasks), the watchdog may also be delayed, as it is not a foreground process in the driver. Driver backlogging or queuing is easily detectable by setting a value in register in the PLC, and then timing how long it takes to receive a feedback via another tag which was scanned to the same register.

7. Appendix A

7.1. Compatibility with Previous FX drivers.



WARNING

MITSUFXS is not directly compatible with MITFXS therefore the need to change the driver name which will prevent problems when upgrading.

- A. X and Y regions scanned as INT and REAL will now be correctly addressed starting bit will now designate the address i.e. X 0 I – 1st integer in the X region X 20 I is the 2nd integer. The old style was X 0 I, X 2 I ...
- B. S regions scanned as INT and REAL will now be correctly addressed starting bit will now designate the address i.e. S 0 I – 1st integer in the S region S 16 I is the 2nd integer. The old style was S 0 I, S 1 I ...

This will now bring the addressing standard into line with GX Developer.

Dword support where applicable, is added.

PDU data lengths have been optimized.

8. Revisions

8.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	25-Jan-2023	K. Robinson	Changes to format, updated image resources.	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

8.2. DLL Revision

Rev.	Date	Changes
4	14/01/2005	Fixes added to D registers below 1024 , 1024-7999 not available. Plc model only allowed even selection...
6	09/12/2005	Fix added to M region bits were not being unpacked correctly.
7	09/12/2005	Added FX3U PLC support.
9	23/10/2009	Added D Register Float Support
14	18/01/2012	Added Modem and Ethernet Support to the Driver
16	08/02/2012	Added Extended FX Protocol Support for FX2N and FX3U to the Driver.
18	14/06/2013	Added test read to dialog
18	01/10/2013	Updated documentation
22	26/09/2018	Fixed driver lockup on ethernet comms error , fixed performance counters not visible when running the Agent Server as a service.

